

DOTNAPPS INVOICE

V1 Product Requirements & Vibe-Coding Specification

Product promise: Create. Quote. Invoice. Share. Get Paid.

Purpose

Build Dotnapps Invoice as a production-grade, multi-tenant SaaS for freelancers, agencies, consultants, startups and SMEs. V1 intentionally excludes AI. The product focuses on reliable quotations, invoicing, GST-aware billing, payments, bulk bill sharing, reminders, expenses, reports, subscriptions and administration.

V1 principles

- No AI in V1; keep clean service boundaries for a future AI layer.
- Desktop-first SaaS UI with full mobile responsiveness.
- Strict organization/business data isolation.
- Quotation → accepted → invoice conversion without duplicate entry.
- Bulk sending is a first-class workflow with filters, preview, progress, failures, retry and history.
- Professional PDF documents and customer-facing document pages are core features.
- Use reusable components and modular services.

1. Product Modules

Module	V1 requirements
Authentication	Sign up, login, logout, email verification, password reset, sessions
Business onboarding	Business profile, logo, GST, numbering, payment details, branding
Dashboard	Revenue, receivables, overdue, expenses, quotation/invoice summaries, activity
Quotations	Create/edit/preview/send, view tracking, accept/reject, expiry, convert to invoice
Invoices	Create/edit/preview/send, statuses, partial payments, overdue, credit/debit notes
Bulk sharing	Multi-select, filters, WhatsApp/email/SMS, reminders, PDF download, export, history
Customers	Profiles, GSTIN, addresses, transaction history, balances
Products & services	Catalog, SKU, HSN/SAC, price, tax, active/inactive
Payments	Full/partial payment, methods, receipts, references
Expenses	Manual entry, categories, tax, receipt upload
GST	CGST/SGST/IGST, HSN/SAC, place of supply, summaries
Reports	Sales, invoices, quotations, payments, receivables, expenses, GST, P&L;
Reminders	Configurable rule-based email/WhatsApp/SMS reminders
Settings	Business, templates, taxes, messaging, team, security, integrations
SaaS billing	Plans, trial, subscriptions, usage limits, upgrades/downgrades
Super admin	Users, businesses, plans, subscriptions, usage, messaging, support, audit

2. Application Architecture

Recommended structure

Frontend: React/Next.js-style TypeScript application with reusable components, forms, tables, charts and responsive layouts.

Backend: API/service layer for authentication, tenant authorization, billing logic, documents, messaging jobs, reminders, reports and audit logs.

Database: relational database with organization-level scoping on every tenant-owned record.

Storage: object storage for logos, PDFs and receipts.

Jobs: background queue for bulk messaging, reminders and PDF batches.

Payments/messaging: provider adapters so vendors can be swapped without changing business logic.

Primary navigation

Dashboard → Quotations → Invoices → Customers → Products & Services → Payments → Expenses → Reports → Reminders → Settings

3. Roles & Permissions

Role	Access
Owner	Full access, billing, business settings, team, integrations
Admin	Operational management and reports
Accountant	Invoices, quotations, payments, expenses, reports
Sales	Customers, quotations, products; restricted financial actions
Staff	Only explicitly granted modules/actions

Backend authorization is the security source of truth; frontend hiding is not sufficient.

4. Authentication & Onboarding

- Signup → email verification → business creation → onboarding → dashboard.
- Login, logout, password reset and session invalidation.
- Business setup: name, owner, contact, address, state, GSTIN/registration type, logo, currency, tax defaults, invoice/quotation numbering, bank/UPI/payment details.
- Do not allow sending a document until required business/document settings are valid.

5. Dashboard

- KPI cards: revenue, outstanding, overdue, expenses, net income.
- Quotation metrics: quoted, accepted, pending, rejected.
- Invoice metrics: draft, sent, partial, paid, overdue.
- Revenue chart with date range.
- Receivables/aging widget.
- Recent invoices, quotations, payments and expenses.
- Quick actions: Create Invoice, Create Quotation, Add Customer, Add Expense.

6. Customer Module

- Searchable/filterable/paginated customer list with export.

- Customer fields: name, company, email, phone, GSTIN, billing/shipping address, state, notes.
- Customer detail: overview, quotations, invoices, payments, outstanding and overdue balances, activity.
- Archive customers with history instead of destructive deletion where transactions exist.

7. Products & Services

- Fields: name, description, SKU, HSN/SAC, unit, default price, tax rate, active/inactive.
- Searchable selector inside quotation/invoice forms.
- Allow line-item price override without altering catalog defaults.
- Centralize tax calculation in a tested service.

8. Quotations — Complete Flow

DRAFT → SENT → VIEWED → ACCEPTED / REJECTED / EXPIRED → ACCEPTED QUOTATION → CONVERT TO INVOICE

- Create from scratch, existing customer or reusable data.
- Auto-generate unique quotation number using business prefix/sequence.
- Quotation date and valid-until date.
- Line items: item, description, quantity, unit, rate, discount, tax, amount.
- Subtotal, discount, taxable value, CGST/SGST/IGST and grand total.
- Notes and terms & conditions.
- Preview, save draft, email/WhatsApp/SMS, secure public link.
- Customer public view supports Accept/Reject.
- Record sent/viewed/accepted/rejected timestamps.
- Accepted quotation converts to invoice and links both records.

Quotation screen

Header → Customer → Items → Totals → Terms → Preview → Save Draft / Send

9. Invoices — Complete Flow

DRAFT → SENT → VIEWED → PARTIALLY PAID → PAID; unpaid invoices become OVERDUE after due date.

- Create from quotation, from scratch or by copying existing data.
- Unique invoice number per business.
- Customer, items, discounts, GST, payment terms, due date, notes, bank/payment details.
- Support standard and GST invoice templates.
- Partial and multiple payments.
- Balance = total – allocated payments.
- Balance zero → PAID. Due date passed with balance > 0 → OVERDUE.
- Cancelled invoices remain auditable.
- Credit/debit notes reference source invoices.

10. PDF & Document Engine

- A4 print-ready PDFs.
- Web preview and PDF must use the same calculation/data layer.
- Templates: Minimal, Modern, Professional, Corporate, GST Classic.
- Business logo, brand accent, typography, footer, bank/UPI details.
- Consistent filenames and secure authorized downloads.
- PDF totals must match application totals.

11. Bulk Bill Share — Core Feature

INVOICES → FILTER/SELECT → BULK ACTION → CHANNEL → MESSAGE PREVIEW → RECIPIENT VALIDATION → CONFIRM → BACKGROUND PROCESSING → RESULTS → BULK ACTIVITY

- Multi-select across paginated results.
- Filters: status, customer, date range, due date, amount range, invoice number.
- Actions: Send Invoice, Send Reminder, Download PDFs, Export, Print, Mark as Paid where allowed.
- Channels: Email, WhatsApp, SMS, subject to provider setup and plan limits.
- Message variables: customer name, invoice number, amount, due date, payment link.
- Before sending show invoice count, total amount, valid recipients, missing contacts and estimated usage.
- Process asynchronously in batches; one failure must not stop the batch.
- Per-item statuses: queued, sent, delivered where supported, failed, skipped.
- Failure reason and retry failed items.
- Bulk Activity stores batch name, creator, time, channel, totals, success/failure/skipped.

Example bulk filters

All overdue invoices | all unpaid invoices due this week | all invoices for Customer X | invoices in date range | invoices above ■50,000

12. Payments & Receipts

- Full or partial payment.
- Fields: invoice, customer, amount, date, method, transaction/reference ID, notes.
- Methods: UPI, bank transfer, card/payment gateway, cash, cheque, other.
- Automatically update balance/status.
- Generate payment receipt.
- Keep payment history auditable; use reversals/adjustments instead of destructive financial edits.

13. Payment Reminders

- Rule-based reminders: 7 days before, 2 days before, due date, 3 days overdue, 7 days overdue, 30 days overdue.
- Enable/disable each rule.
- Editable templates.
- Channels: email, WhatsApp, SMS.
- Record reminder events to prevent duplicate sends.
- Manual reminder and bulk reminder actions.

14. Expenses

- Manual expense: vendor, amount, date, category, payment method, tax, notes, receipt.
- Receipt upload as image/PDF.
- Configurable categories.
- Expense data feeds reports.

15. GST / India-First Billing

- GSTIN and registration type.
- HSN/SAC, place of supply, taxable value.
- CGST, SGST, IGST.
- Tax-inclusive and tax-exclusive pricing.
- Centralized tested calculation service.
- Configurable compliance rules because regulations can change.

Production note: validate GST calculations, document formats and future e-invoicing/e-way-bill integrations against current Indian requirements before launch.

16. Reports

- Sales by period/customer/product.
- Invoice paid/unpaid/overdue/cancelled.
- Quotation accepted/rejected/expired and conversion rate.
- Payment collections by period/method/customer.
- Receivables aging: current, 1–30, 31–60, 61–90, 90+ days.
- Expenses by category/vendor/period.
- GST summary.
- Profit & Loss.
- Date filters and CSV/XLSX export.

17. Client-Facing Document Portal

- Secure invoice/quotation viewing link.
- Customer can download PDF, accept/reject quotation, and pay invoice when gateway is enabled.
- Business branding.
- Track viewed/accepted timestamps.
- No unrelated tenant/customer data exposure.

18. Settings

- Business Profile; Invoice Settings; Quotation Settings; GST & Tax; Payment Settings; Templates; Email; WhatsApp; SMS; Reminder Settings; Team & Permissions; Notifications; Integrations; Subscription & Billing; Security & Sessions.

19. SaaS Subscription Architecture

- Free trial → plan selection → payment → active subscription.
- Configurable plans and limits.
- Meter invoices, quotations, customers, team members, bulk sends, messages and storage as needed.
- Server-side enforcement.
- Immediate upgrade behavior; safe downgrade behavior.

- Failed payment → retry/grace period → possible suspension.
- Never delete business data because of subscription status.

20. Super Admin Panel

- Dashboard: users, businesses, active subscriptions, MRR, invoice volume, quotation volume, payment volume, messaging usage.
- Users and businesses search/detail.
- Plans and limits.
- Subscriptions and failed payments.
- Messaging usage and provider status.
- Support.
- System settings.
- Privileged admin audit log.

21. Core Data Model

User, Organization/Business, Membership, Role, Permission, Customer, CustomerAddress, ProductService, Quotation, QuotationItem, Invoice, InvoiceItem, CreditNote, DebitNote, Payment, PaymentAllocation, Expense, ExpenseAttachment, DocumentTemplate, ReminderRule, ReminderEvent, MessageTemplate, BulkSendBatch, BulkSendItem, SubscriptionPlan, Subscription, UsageRecord, AuditLog, Notification, Integration

Database rules

- Every tenant-owned table has organization_id/business_id and every query is tenant-scoped.
- Unique quotation/invoice numbers are scoped to the business.
- Index organization_id, customer_id, invoice_id, quotation_id, status, due_date and created_at.
- Use transactions for quotation conversion, payments and status transitions.
- Financial history should be auditable.

22. API / Service Areas

/auth /businesses /customers /products-services /quotations /invoices /payments /expenses /reports /reminders /bulk-messaging /documents /subscriptions /settings /admin

- Consistent API response/error format.
- Authorization on every protected mutation.
- Long-running bulk operations return job/batch IDs.
- Use idempotency for payment/webhook-sensitive operations.

23. Webhooks & Integrations

- Verify payment webhook signatures and make handlers idempotent.
- Capture delivery/bounce/failure events where providers support them.
- Keep payment and messaging vendors behind adapters/interfaces.

24. UX / UI Requirements

- Modern, minimal, trustworthy SaaS aesthetic; avoid legacy accounting-software feel.
- Desktop sidebar + responsive mobile navigation.
- Clear status badges.
- Search, filter, sort, pagination and useful table actions.
- Fast invoice/quotation creation.
- Inline validation and clear error messages.
- Designed loading, empty, success and error states.
- Confirm destructive actions.
- Consistent currency/date formatting.
- Keyboard-friendly and accessible controls.

25. Primary Screens

Landing → Pricing → Login → Signup → Onboarding → Dashboard → Quotation List → Create/Edit Quotation → Quotation Preview/Public View → Invoice List → Create/Edit Invoice → Invoice Preview/Public View → Bulk Send → Bulk Activity → Customer List → Customer Detail → Product/Service List → Payments → Payment Receipt → Expenses → Reports → Reminders → Templates → Team & Permissions → Integrations → Subscription/Billing → Security → Admin screens

26. Acceptance Criteria

- New business can sign up, onboard and create its first quotation.
- Accepted quotation converts to invoice with preserved data.
- Invoice preview, PDF and send flow works.
- Partial payments correctly update balance/status.
- Overdue status is deterministic and safe against stale jobs.
- Bulk send works across filtered/paginated selections.
- One failed recipient does not stop the batch.
- Failed items can be retried.
- Reminder rules do not duplicate events.
- Tenant A cannot access Tenant B data/files/reports.
- Financial mutations have audit trails.
- Subscription limits are server-enforced.
- GST calculations are covered by automated tests.

27. Testing Strategy

- Unit tests: tax, totals, discounts, payment allocation, status transitions.
- Integration tests: quotation conversion, invoice, payment, reminders, bulk jobs.
- Authorization/cross-tenant tests.
- End-to-end: signup → onboarding → quotation → acceptance → invoice → payment.
- Bulk tests with small and large batches, mixed failures and retries.

- PDF visual/snapshot checks.
- Webhook idempotency tests.
- Subscription limit tests.
- Accessibility checks.

28. Security & Reliability

- Strong password hashing; secure session strategy.
- Server-side validation and authorization.
- Rate limiting on sensitive endpoints.
- Secure private file access.
- Secrets stored securely and never committed.
- Database backups and restore process.
- Audit logging.
- Structured logging, error tracking and health checks.

29. Recommended Build Order

Phase 1 – Foundation: Auth → organization model → roles → database → design system → navigation → settings

Phase 2 – Core billing: Customers → products/services → quotations → quotation PDF → invoices → invoice PDF

Phase 3 – Collections: Payments → partial payments → receipts → statuses → reminders

Phase 4 – Bulk: Multi-select → filters → async sending → retries → bulk history → export/download

Phase 5 – Finance: Expenses → reports → GST summaries

Phase 6 – SaaS: Plans → subscriptions → usage limits → super admin

Phase 7 – Hardening: Security → tests → performance → accessibility → monitoring → production readiness

30. Explicitly Exclude from V1

- AI assistant or AI-generated documents.
- Complex full accounting/ledger replacement.
- Payroll.
- Inventory/warehouse management.
- ERP functionality.
- Overly broad compliance automation before core billing is stable.

31. Future V2

- AI assistant and natural-language actions.
- AI receipt/document extraction.
- Smart collection insights.
- Recurring billing automation.
- Advanced client portal.
- Inventory/purchases.
- Deeper accounting integrations.

- Applicable e-invoicing/e-way-bill integrations.
- Mobile apps.

32. Master Vibe-Coding Instruction

Build Dotnapps Invoice as a production-grade multi-tenant SaaS, not a static demo. Use this document as the source of truth. Do not add AI in V1. Implement real persistence, authentication, tenant isolation, roles, quotation/invoice lifecycles, quotation-to-invoice conversion, professional PDFs, payment tracking, bulk bill sharing with asynchronous jobs and per-item results, reminders, GST-aware calculations, reports, subscriptions and a separate super-admin panel. Do not fake backend behavior with local-only state. If an integration is not configured, show a proper setup state rather than pretending a message or payment was processed. Use reusable components, validation, loading/empty/error states, audit logs, tests and secure file handling. Keep modules loosely coupled so AI can be added later without rewriting the billing core.

33. Definition of Done

- All V1 screens responsive and functional.
- Database persistence complete.
- Authentication, authorization and tenant isolation verified.
- Quotation and invoice lifecycles complete.
- Quotation-to-invoice conversion complete.
- PDFs match on-screen totals.
- Bulk sending supports failures/retries and history.
- Payments/reminders tested.
- GST calculations tested.
- Reports reconcile with transactions.
- Subscription limits enforced.
- Admin panel operational.
- Security/accessibility/observability review completed.